

Vision-Based 3D Fruit Localization for Robotic Harvesting Using Recursive State Estimation

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Abstract—Precision agriculture is increasingly reliant on autonomous robotic systems with advanced perception capabilities for tasks such as yield estimation and automated harvesting. However, accurate 3D localization of crops, particularly fragile fruits that are prone to bruising and require gentle handling, remains challenging due to occlusions, variable lighting, and dynamic field conditions. To address this, we propose a novel framework for fruit 3D localization that leverages temporal associations between consecutive video frames, and employs a Recursive Least Squares (RLS) estimator. This approach enables robust and precise position tracking of individual fruits, facilitating both non-invasive yield prediction and targeted harvesting operations. The proposed algorithm was extensively validated in both simulated environments and real-world scenarios using an autonomously mobile robot platform equipped with multi-sensor fusion. Experimental results demonstrate the system's effectiveness in improving localization accuracy and reliability, offering a practical solution for precision agriculture applications. This research contributes to bridging the technological gap in agricultural robotics, particularly in enhancing 3D perception for fruit monitoring and harvesting.

Index Terms—Precision Agriculture, 3D Fruit Localization, Recursive Least Squares, Autonomous Robotics, Multi-sensor Fusion

I. INTRODUCTION

Precision agriculture technology, as a key driver in enhancing the food quality, efficiency, and sustainable development of the food industry, is significantly improving the

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standardization and refinement of crop management and food processing through the integration of advanced technologies such as machine vision, automatic control, and intelligent robotics [1]. Currently, autonomous robotic systems equipped with multimodal perception and intelligent decision-making capabilities are progressively being applied in critical stages such as crop growth monitoring [2], food quality inspection, sorting and grading, aseptic packaging, contaminant detection, and full-chain traceability [3]. These advancements continue to steer food production models toward high precision, flexibility, and intelligent transformation.

With the rapid advancement of artificial intelligence, robotics, and Unmanned Aerial Vehicle (UAV) technology, as these cutting-edge tools permeate diverse sectors, their inherent automation capabilities are profoundly reshaping a multitude of industries [4]. As one of the oldest industries, agriculture is also undergoing constant transformation. In agricultural production, the rational arrangement of production plans and resource allocation relies on accurate predictions of crop yields. Therefore, precisely estimating the yield of a specific plot has become a critical issue in agricultural management [5]. To obtain average yield data for an entire province or even the entire country, sampling survey methods are generally employed, where the overall yield is extrapolated from sample data [6]. Among these methods, "field cutting and measuring" is a common on-site survey technique. In recent years, related research directions such as "biomass estimation based on tree counting and mapping," "orchard fruit counting

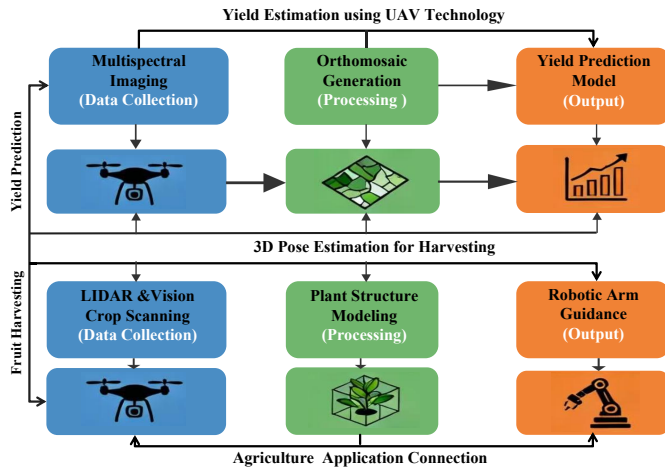


Fig. 1. Critical tasks in autonomous intelligent agriculture.

and yield estimation,” and “autonomous sampling and environmental modeling for precision forestry and agriculture” have all been dedicated to achieving this goal. As shown in Fig. 1, we identify two critical tasks in autonomous intelligent agriculture: yield estimation and harvesting operation.

In terms of yield estimation, neural networks can be trained using historical data for preliminary predictions, and the accuracy of the results can be further improved by incorporating localized sampling data from the current year [7]. In practical implementation, different crops require distinct sampling strategies. For example, in rice production, localized sampling and field-cutting measurements can be performed before full maturity. In contrast, for fruit crops, direct sampling before complete ripening is often challenging, making methods reliant on counting and estimation more common [8]. It is important to note that although “counting” and “estimation” share the same objective, they differ significantly in methodology. “Counting” typically refers to enumerating individual fruits one by one to calculate the total yield. However, in real-world agricultural settings, due to constraints in time and cost, indirect methods such as sampling or density estimation are often used for approximate assessments rather than comprehensive enumeration [9].

The next critical phase after yield estimation is the harvesting operation. For field crops such as rice, combine harvesters have achieved mature automated operations. However, in horticultural crop scenarios like fruit trees, the picking process requires careful handling to avoid damaging the plants, placing higher demands on operational precision—a domain where current technologies remain underdeveloped. Although sporadic studies have explored this field, precise 3D perception technology in agricultural environments is still in its early stages compared to advanced industries such as autonomous driving. To address these bottlenecks, this paper focuses on the aforementioned challenges, with main contributions encompassing the following two aspects:

1) We introduce a framework for estimating the 3D localization of fruits by utilizing associations between consecutive

frames based on a recursive least squares estimator.

2) The algorithm was validated in both simulated and real-world scenarios by an autonomously mobile robot platform with integrated multi-sensor fusion.

II. RELATED WORK

In the field of crop yield calculation, biomass estimation is a relatively established domain with broad applications in forestry, animal husbandry, and agriculture. This approach shares similarities with statistical sampling, following a fundamental two-step process: first, collecting samples, and then processing the sample data to extrapolate estimates to larger scales, such as for individual trees or forest stands [10]. However, fruit yield estimation requires sampling the current year’s fruit set rate, yet practical calculation of this rate still faces numerous challenges. These include determining the number of fruit-bearing mother branches, counting flowers on these branches, quantifying fruit set after blossom drop, and precisely locating fruits—tasks that are both labor-intensive and prone to error when done manually. The advancement of AI [11] and Computer Vision (CV) [12] has introduced innovative solutions to address these difficulties.

The most common application scenarios for 3D object detection are in autonomous driving and automated sorting systems. In autonomous driving, it is typically necessary to estimate surrounding object positions, predict motion, and adjust trajectory planning accordingly. Well-known benchmarks include the KITTI 3D Object Detection benchmark [13], among others. Existing solutions include camera-based methods, LiDAR-based techniques, sensor fusion networks [14], and Bird’s Eye View (BEV) representations [15].

However, in the agricultural sector, developments in datasets, benchmarks, hardware, and algorithms lag behind those in autonomous driving and robotics [5]. In terms of datasets and benchmarks, authoritative standards are still lacking. For hardware, RGB-D cameras or combinations of cameras and LiDAR are primarily used. At the algorithmic level, early research mostly focused on fruit-picking scenarios, estimating the position and status of individual fruits (such as whether the fruit is occluded or intact) to drive robotic arms for grasping tasks. However, approaches that only estimate the position of individual fruits do not fully align with actual production needs. Recently, macro-estimation studies such as orchard-level fruit counting have been gradually emerging.

None of these methods is entirely accurate, as errors are always inevitable. Sensor inaccuracies, localization errors, and 3D algorithm limitations lead to either duplicated detections or missed detections. Techniques such as loop closure and scene graphs have been applied to mitigate these errors, but no approach has yet systematically analyzed the proportion of errors contributed by different modules. Therefore, this paper experimentally examines the error sources associated with various sensor combinations and proposes an Extended Kalman Filter (EKF) [16] method to enhance the accuracy of fruit tracking. In summary, the contributions of this paper are as follows:

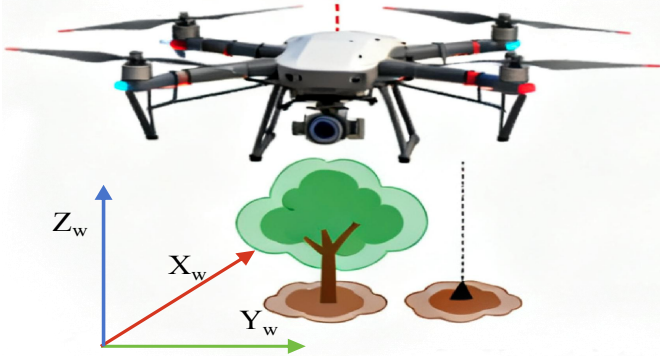


Fig. 2. Uncertainty and its propagation in 3D object estimation pipeline.

1) An EKF-based algorithm for estimating the 3D position of fruits by leveraging inter-frame associations is proposed.

2) The algorithm is validated using an autonomous mobile robot platform with multi-sensor fusion in both simulated and real-world scenarios.

III. METHODOLOGY

A naive approach for 3D object pose estimation involves first detecting and localizing objects in the image (e.g., using methods like YOLO), and then combining the detection results with depth information to estimate the object's position in the world frame. However, as depicted in Fig. 2, in practice due to computational errors, the estimated position of the object in the camera frame often exhibits jittery trajectories with significant fluctuations. Moreover, in outdoor environments, the vehicle's own position—typically estimated via odometry—also introduces errors, which further propagate to the object's computed location in the world coordinate system. To address these issues, backend optimization methods are required to refine the initially coarse 3D object poses.

Our methodology is divided into two main parts: the first introduces the sensor setup used to compute the 3D positions of objects, and the second presents the backend optimization algorithm. Before proceeding, we first define the mathematical notations used throughout this paper, including world coordinate systems, Lie groups, and Lie algebras.

We represent a 3D rotation matrix by $\mathbf{R} \in \mathbb{R}^{3 \times 3}$, a 3D translation vector by $\mathbf{t} \in \mathbb{R}^3$, and a 3D rigid body transformation by $SE(3) = \{\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} \in \mathbb{R}^{4 \times 4} | \mathbf{R} \in SO(3), \mathbf{t} \in \mathbb{R}^3\}$. Given two transformation matrices $\mathbf{T}_1$ and $\mathbf{T}_2$. We have $\{\mathbf{T}_1 = \begin{bmatrix} \mathbf{R}_1 & \mathbf{t}_1 \\ \mathbf{0} & 1 \end{bmatrix} \in \mathbb{R}^{4 \times 4} | \mathbf{R}_1 \in SO(3), \mathbf{t}_1 \in \mathbb{R}^3\}$ and $\{\mathbf{T}_2 = \begin{bmatrix} \mathbf{R}_2 & \mathbf{t}_2 \\ \mathbf{0} & 1 \end{bmatrix} \in \mathbb{R}^{4 \times 4} | \mathbf{R}_2 \in SO(3), \mathbf{t}_2 \in \mathbb{R}^3\}$ in $SE(3)$ —the Special Euclidean Group in three dimensions—their composition is defined as $\mathbf{T}_1 \cdot \mathbf{T}_2 = \begin{bmatrix} \mathbf{R}_1 \mathbf{R}_2 & \mathbf{t}_1 + \mathbf{R}_1 \mathbf{t}_2 \\ \mathbf{0} & 1 \end{bmatrix}$. The notation T_{21} is used to indicate the transformation from coordinate frame 1 to coordinate frame 2.

In a particular application, we wish to sum two vectors: r_v^{pq} and r_w^i . The vector r_w^i extends from point q to point i and is expressed in the world coordinate system, whereas r_v^{pq}

goes from point p to point q and is represented in the vehicle coordinate frame. To perform the addition, we first convert r_v^{pq} into the world coordinate system via the transformation T_{wv} . The resulting vector is then added to r_w^i . This procedure is summarized by the following expression:

$$r_w^{pi} = T_{wv} r_v^{pq} + r_w^i. \quad (1)$$

Let f denote the focal length in meters, while α and β represent the number of pixels per meter along the x and y axes respectively. Consequently, $f_x = \alpha f$ and $f_y = \beta f$, together with c_x and c_y , are all expressed in pixels. This relationship can be more compactly represented in matrix form, though it requires using homogeneous coordinates on the left side and non-homogeneous coordinates on the right:

$$\begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \frac{1}{Z} \begin{pmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \frac{1}{Z} \mathbf{K} \mathbf{P}. \quad (2)$$

Alternatively, multiplying both sides by Z yields the more conventional form found in most literature:

$$Z \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \begin{pmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \mathbf{K} \mathbf{P}. \quad (3)$$

In these equations, the middle matrix $\mathbf{K}$ is referred to as the camera's intrinsic parameter matrix (or simply intrinsics). These parameters are typically fixed after camera manufacturing and remain constant during operation. While some manufacturers provide these intrinsic values, they often need to be estimated through a process known as camera calibration. Established calibration methods, such as the widely-used approach [15], have reached sufficient maturity that detailed discussion here becomes unnecessary.

A. Object Detection Frontend

The frontend perception pipeline operates on a per-frame basis. It first computes the position of detected objects relative to the robot platform using instantaneous sensor readings. This local estimate is then converted into a global position within the world coordinate system, utilizing the robot's estimated pose provided by its odometry. In terms of sensor configuration, we investigate two primary approaches: one using a standalone RGB-D camera, and another that employs fusion between a camera and a lidar. The 3D reconstruction is achieved by back-projecting the 2D image detections into 3D space using the camera's projective geometry (as defined in (2) and the associated depth measurements from the sensor).

1) *RGB-D camera 3D object detection*: RGB-D cameras, which provide depth information, can be primarily classified into stereo-based and Time-of-Flight (ToF) technologies. For readers interested in detailed principles, we recommend the work by Kang *et al.* [6]. Regardless of the specific technology, RGB-D cameras generally have a limited operating range and are sensitive to environmental illumination conditions, making them more suitable for indoor applications.

A practical advantage of these sensors is that depth values are directly registered to image pixels. This allows us to conveniently extract depth information from within a target object's bounding box to compute its 3D position in the camera coordinate system. Given a target pixel (u, v) with measured depth d , the 3D position (x_c, y_c, z_c) in the camera frame can be computed as:

$$\begin{pmatrix} x_c \\ y_c \\ z_c \end{pmatrix} = \mathbf{K}^{-1} \cdot \begin{pmatrix} u \cdot d \\ v \cdot d \\ d \end{pmatrix}, \quad (4)$$

where $\mathbf{K}$ represents the camera intrinsic matrix as in 2, and the z -axis corresponds to the optical axis (depth direction).

2) *Camera-LiDAR fusion*: The camera-LiDAR fusion approach similarly begins by detecting objects and extracting their bounding boxes from the image. Depth information is then obtained from the LiDAR point cloud. Due to its higher power, LiDAR is less sensitive to illumination changes and offers a longer detection range, making it more suitable for outdoor environments. Several fusion strategies can be employed: the point cloud from the current frame can be projected onto the image plane; candidate points can be selected using a camera field-of-view (FOV) cone within the point cloud; or depth can be retrieved from a pre-built LiDAR prior map. However, due to the relatively low resolution of LiDAR, larger errors may occur when detecting small objects such as fruits. Therefore, in this study, this approach is implemented as a comparative baseline.

The first step in LiDAR-camera fusion is extrinsic calibration. Assuming the camera intrinsics are known, extrinsic calibration methods can be divided into target-based and targetless approaches. Target-based methods require a calibration target of known dimensions, typically a checkerboard. With relatively fixed positioning, an initial extrinsic transformation between the LiDAR and camera can be estimated from CAD data. To collect data for LiDAR-camera calibration, the LiDAR is mounted on a tripod and synchronized with the camera to capture simultaneous RGB images and point clouds. Each scene is recorded for 15 seconds to accumulate a sufficient number of point cloud frames.

The two calibration methods differ in setup. For the target-based approach, a checkerboard with 7×5 inner corners and a grid size of 5 cm is used. The board is placed in various positions and orientations within the overlapping FOV. Careful selection and placement of the calibration target are essential for successful calibration. For the targetless method, scenes rich in geometric features are chosen. This approach imposes specific requirements on scene structure: for example, scenes containing purely cylindrical objects such as pillars are unsuitable, as edge extraction relies on plane fitting. Additionally, extracted edges should be distributed relatively uniformly across the FOV to reduce measurement noise, and should exhibit diversity in orientation rather than being predominantly aligned in one direction (e.g., vertical).

The initial extrinsic matrix is obtained from CAD models and further refined through edge-based matching. We refer

reader to [14] an example of LiDAR-camera extrinsic calibration using a targetless method, showing the original RGB image, extracted features, depth projection on the RGB image, and the colored point cloud. The resulting extrinsic matrix is recorded for subsequent accuracy evaluation. However, since this is not the main focus of the paper, no quantitative evaluation is provided. Finally, the 3D position of the object can be reconstructed based on the derived depth information as in (4).

B. Object Pose Backend Optimization

Let the object's state vector be defined as:

$$\mathbf{x} = [x, y, z, \phi, \theta, \psi]^T, \quad (5)$$

where (x, y, z) represent the 3D position coordinates and (ϕ, θ, ψ) represent the roll, pitch, and yaw Euler angles.

The measurement model at time k is:

$$\mathbf{z}(k) = \mathbf{H}(k)\mathbf{x} + \mathbf{w}(k), \quad (6)$$

where $\mathbf{H}(k)$ is the observation matrix and $\mathbf{w}(k) \sim \mathcal{N}(0, \mathbf{R}(k))$ is zero-mean Gaussian measurement noise.

1) *Batch Least Squares Formulation*: This section presents the derivation of the batch optimization approach for backend processing, where all captured frames are processed simultaneously to perform offline refinement of the target object's estimated position. The batch least squares estimate with k measurements is given by:

$$\hat{\mathbf{x}}(k) = [\mathbf{H}^{kT} \mathbf{R}^{k-1} \mathbf{H}^k]^{-1} \mathbf{H}^{kT} (\mathbf{R}^k)^{-1} \mathbf{z}^k, \quad (7)$$

with error covariance:

$$\mathbf{P}(k) = [\mathbf{H}^{kT} \mathbf{R}^{k-1} \mathbf{H}^k]^{-1}. \quad (8)$$

2) *Recursive Formulation*: While the batch formulation provides an optimized solution for least squares problems, it is not feasible for online systems such as robotics tasks—including fruit picking and manipulation—where real-time output is required. We therefore propose a recursive approach to estimate the position of the target object. When a new measurement $\mathbf{z}(k+1)$ arrives, the information matrix updates as:

$$\mathbf{P}(k+1)^{-1} = \mathbf{P}(k)^{-1} + \mathbf{H}(k+1)^T \mathbf{R}(k+1)^{-1} \mathbf{H}(k+1). \quad (9)$$

Using the matrix inversion lemma, the covariance update becomes:

$$\mathbf{P}(k+1) = [\mathbf{I} - \mathbf{W}(k+1)\mathbf{H}(k+1)] \mathbf{P}(k), \quad (10)$$

where the Kalman gain $\mathbf{W}(k+1)$ is defined as:

$$\mathbf{W}(k+1) = \mathbf{P}(k)\mathbf{H}(k+1)^T \mathbf{S}(k+1)^{-1}, \quad (11)$$

and the residual covariance is:

$$\mathbf{S}(k+1) = \mathbf{H}(k+1)\mathbf{P}(k)\mathbf{H}(k+1)^T + \mathbf{R}(k+1). \quad (12)$$

The state estimate updates recursively as:

$$\hat{\mathbf{x}}(k+1) = \hat{\mathbf{x}}(k) + \mathbf{W}(k+1) [\mathbf{z}(k+1) - \mathbf{H}(k+1)\hat{\mathbf{x}}(k)]. \quad (13)$$

C. Observation Model for 3D Pose

For RGB-D camera-based 3D pose estimation, the observation matrix $\mathbf{H}(k)$ relates the object's 6DOF pose to the measured point cloud features. This linear relationship allows us to connect the state vector to the actual sensor measurements. For position measurements derived from depth data, the observation matrix takes the form:

$$\mathbf{H}_{\text{pos}}(k) = [\mathbf{I}_{3 \times 3} \quad \mathbf{0}_{3 \times 3}] \quad (14)$$

This formulation directly extracts the positional components from the state vector while ignoring orientation.

Similarly, for orientation measurements obtained from visual features in the RGB image, the observation matrix becomes:

$$\mathbf{H}_{\text{ori}}(k) = [\mathbf{0}_{3 \times 3} \quad \mathbf{I}_{3 \times 3}] \quad (15)$$

This structure isolates the orientation components from the full state vector.

1) *Measurement Noise Covariance:* Building upon the observation model, we must characterize the uncertainty associated with each measurement. The measurement noise covariance $\mathbf{R}(k)$ depends on the RGB-D sensor characteristics and varies across different operating conditions:

$$\mathbf{R}(k) = \text{diag}(\sigma_x^2, \sigma_y^2, \sigma_z^2, \sigma_\phi^2, \sigma_\theta^2, \sigma_\psi^2), \quad (16)$$

where the variances are determined through sensor calibration and account for feature detection uncertainty. The diagonal structure assumes independence between different measurement components, which is a reasonable approximation for most practical applications.

2) *Algorithm Implementation:* With the observation model and noise characteristics defined, we now present the complete recursive estimation algorithm that integrates these components into a unified framework for real-time pose estimation.

Algorithm 1 Recursive Least Squares for 3D Pose Estimation

Require: Initial state estimate $\hat{\mathbf{x}}(0)$, initial covariance $\mathbf{P}(0)$

Ensure: Refined state estimates $\hat{\mathbf{x}}(k)$ for each frame

- 1: Initialize: $\hat{\mathbf{x}} \leftarrow \hat{\mathbf{x}}(0)$, $\mathbf{P} \leftarrow \mathbf{P}(0)$
 - 2: **for** each new RGB-D frame $k = 1, 2, \dots$ **do**
 - 3: Extract 3D position and orientation measurements $\mathbf{z}(k)$
 - 4: Compute observation matrix $\mathbf{H}(k)$ based on current features
 - 5: Calculate residual covariance: $\mathbf{S}(k) = \mathbf{H}(k)\mathbf{P}\mathbf{H}(k)^T + \mathbf{R}(k)$
 - 6: Compute Kalman gain: $\mathbf{W}(k) = \mathbf{P}\mathbf{H}(k)^T\mathbf{S}(k)^{-1}$
 - 7: Update state estimate: $\hat{\mathbf{x}} \leftarrow \hat{\mathbf{x}} + \mathbf{W}(k)[\mathbf{z}(k) - \mathbf{H}(k)\hat{\mathbf{x}}]$
 - 8: Update error covariance: $\mathbf{P} \leftarrow [\mathbf{I} - \mathbf{W}(k)\mathbf{H}(k)]\mathbf{P}$
 - 9: Output current estimate: $\hat{\mathbf{x}}(k) = \hat{\mathbf{x}}$
 - 10: **end for**
-

This algorithm efficiently combines the theoretical models with practical implementation considerations, enabling robust and computationally efficient pose estimation suitable for real-time robotic applications.

IV. EXPERIMENTS

In accordance with the error analysis presented in Section III-A, we performed experiments with two distinct sensor configurations: RGB-D and camera-LiDAR fusion. This section provides a quantitative assessment by comparing the algorithm's performance in fruit counting and object localization against the Ground Truth (GT). Counting accuracy is expressed as a percentage, with 100% denoting a perfect match to GT. Deviations from this ideal value signify inaccuracies, where the extent of the deviation is inversely proportional to the algorithm's precision. Gazebo simulation environment is shown in Fig 3.



Fig. 3. Simulation environment and real-world UAV model.

In the Gazebo environment, various fruits are placed on vertical plant beds within the simulated garden. These crops possess similar color and size characteristics. An UAV, equipped with a depth camera and an Inertial Measurement Unit (IMU), performs plant position estimation within the greenhouse. For quantitative evaluation, the ground truth 6-DoF trajectories are obtained directly from the simulation engine with sub-millimeter accuracy, which serves as the absolute reference for computing localization errors. The onboard IMU and RGB-D camera provide only sensor measurements to the algorithm.

Based on the sensor data and the methodology described previously, we calculate the 3D position of the fruits. This calculation is performed for each frame. The data from different frames in the time series are then compared. The sequence from left to right corresponds to the UAV's flight timeline, showing the position estimates for different plants.

We conducted a comparative analysis of estimation accuracy between our method and a baseline moving-average approach applied across two consecutive frames. Both methods utilize the same front-end framework, which employs an RGB-D camera as its input source and relies on ground-truth semantic segmentation annotations. Within this idealized simulation framework, precise semantic segmentation labels are systematically generated using HSV-based thresholding techniques, effectively mitigating sensor noise artifacts.

The results are illustrated in Fig. 4. The green line represents the ground-truth plant position, the red dashed line depicts the trajectory from our proposed method, and the blue solid line shows the result from the moving-average-based method. It

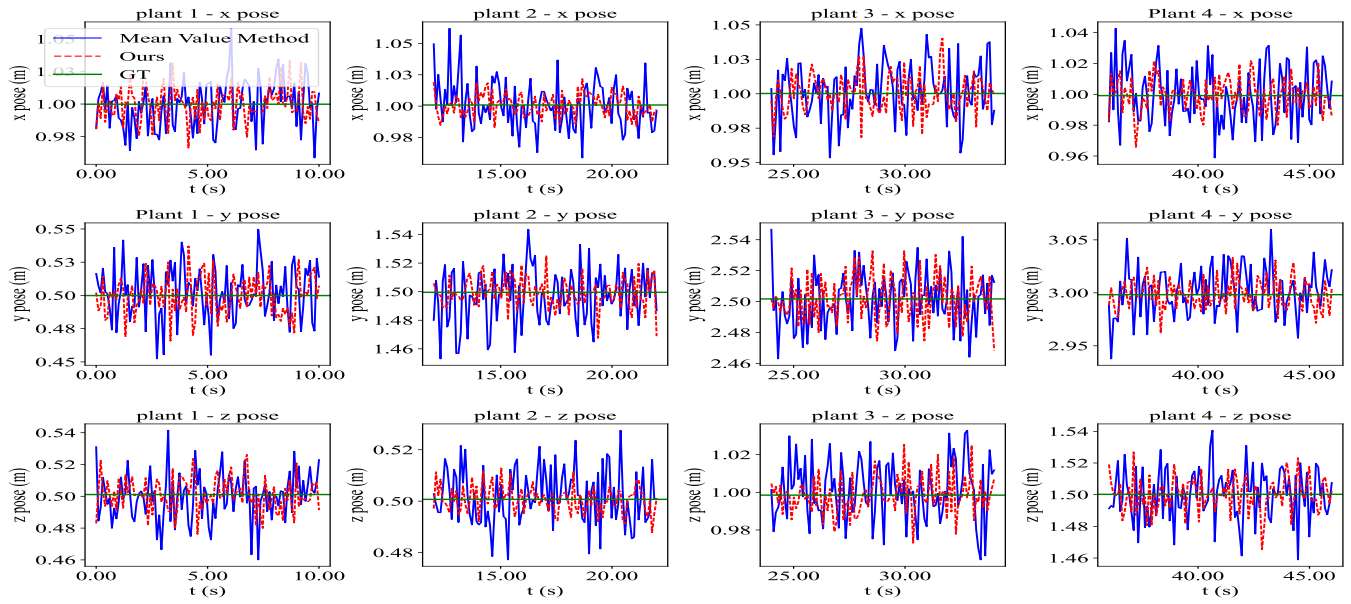


Fig. 4. Plant pose estimation accuracy comparison between our method and the mean value method.

can be observed that both methods are susceptible to inaccuracies in the front-end calculations. However, our method demonstrates a significantly lower variance than the moving-average approach.

V. CONCLUSION

This paper addresses temporal consistency in fruit 3D localization for precision agriculture via a novel framework. Leveraging inter-frame correlations and a recursive least-squares estimator, it robustly tracks fruit positions. Using RLS filters, it enables non-invasive yield prediction and targeted harvesting. The proposed method is evaluated on an autonomous mobile robot with multi-sensor fusion, the system boosts localization accuracy and reliability, outperforming mean-based methods in precision and trajectory coherence. This work delivers a practical solution, closing technical gaps in agricultural robotics, especially for 3D perception. Future work will focus on UAV specific optimizations: reducing latency for lightweight UAV localization, integrating path planning for autonomous patrol/harvesting. This work is currently validated in simulation environments; future work will extend to real-world field experiments with a focus on improving robustness under illumination variations and occlusions in complex agricultural scenarios.

REFERENCES

- [1] UN Secretary-General, "Progress towards the sustainable development goals: Report of the secretary-general," New York, NY, USA, 2024. [Online]. Available: <https://un.org/sdgs/reports>
- [2] C. Smitt, M. Halstead, T. Zaenker, M. Bennewitz, and C. McCool, "Pathobot: A robot for glasshouse crop phenotyping and intervention," in *IEEE Int. Conf. Robot. Autom. (ICRA)*. IEEE, 2021, pp. 2324–2330.
- [3] M. Raj, S. Gupta, V. Chamola, A. Elhence, T. Garg, M. Atiquzzaman, and D. Niyato, "A survey on the role of internet of things for adopting and promoting agriculture 4.0," *J. Netw. Comput. Appl.*, vol. 187, p. 103107, 2021.
- [4] K.-W. Tse, R. Pi, W. Yang, X. Yu, and C.-Y. Wen, "Advancing uav-based inspection system: The ussa-net segmentation approach to crack quantification," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–14, 2024.
- [5] W. Yang, K. Liu, Z. Tan, L.-Y. Lo, Y. Wang, K.-H. Wong, and C.-Y. Wen, "Hierarchical 3-d scene-graph-based semantic-metric slam for plant inspection and fruit counting in intelligent hydroponics system," *IEEE Internet Things J.*, vol. 12, no. 23, pp. 49 071–49 082, 2025.
- [6] H. Kang, X. Wang, and C. Chen, "Accurate fruit localisation using high resolution lidar-camera fusion and instance segmentation," *Comput. Electron. Agric.*, vol. 203, p. 107450, 2022.
- [7] T. Liu, H. Kang, and C. Chen, "Orb-livox: A real-time dynamic system for fruit detection and localization," *Comput. Electron. Agric.*, vol. 209, p. 107834, 2023.
- [8] C. Friess, V. Niclescu, T. Polonelli, M. Magno, and L. Benini, "Fully onboard slam for distributed mapping with a swarm of nano-drones," *IEEE Internet Things J.*, 2024.
- [9] C. Cadena, L. Carlone, H. Carrillo, Y. Latif, D. Scaramuzza, J. Neira, I. Reid, and J. J. Leonard, "Past, present, and future of simultaneous localization and mapping: Toward the robust-perception age," *IEEE Trans. Robot.*, vol. 32, no. 6, pp. 1309–1332, 2016.
- [10] W. Yang, H. Hu, K.-w. Tse, S. Chen, W. Wen, and C.-y. Wen, "Lidar stereo visual inertial pose estimation based on feedforward and feedbacks," in *Int. Conf. Unmanned Aircraft Syst. (ICUAS)*. IEEE, 2024, pp. 1042–1049.
- [11] I. Armeni, Z.-Y. He, J. Gwak, A. R. Zamir, M. Fischer, J. Malik, and S. Savarese, "3d scene graph: A structure for unified semantics, 3d space, and camera," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 5664–5673.
- [12] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in *Adv. Neural Inf. Process. Syst.*, 2012, pp. 1097–1105.
- [13] A. Geiger, P. Lenz, C. Stiller, and R. Urtasun, "Vision meets robotics: The kitti dataset," *Int. J. Robot. Res.*, vol. 32, no. 11, pp. 1231–1237, 2013.
- [14] W. Yang, H. Luo, K.-W. Tse, H. Hu, K. Liu, B. Li, and C.-Y. Wen, "Autonomous targetless extrinsic calibration of thermal, rgb, and lidar sensors," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–11, 2024.
- [15] T. Qin, P. Li, and S. Shen, "Vins-mono: A robust and versatile monocular visual-inertial state estimator," *IEEE Trans. Robot.*, vol. 34, no. 4, pp. 1004–1020, 2018.
- [16] G. P. Huang, A. I. Mourikis, and S. I. Roumeliotis, "Observability-based rules for designing consistent ekf slam estimators," *Int. J. Robot. Res.*, vol. 29, no. 5, pp. 502–528, 2010.